

APPLICATIONS OF DIFFERENTIATION

Dr. Gabriel Obed Fosu

Department of Mathematics

Kwame Nkrumah University of Science and Technology

Google Scholar: <https://scholar.google.com/citations?user=ZJfCMyQAAAAJ&hl=en&oi=ao>

ResearchGate ID: https://www.researchgate.net/profile/Gabriel_Fosu2



Lecture Outline

- 1 L'hospital's Theorem
- 2 Critical Values
- 3 Minimum and Maximum Values
- 4 Mean value theorems
- 5 Shapes of graphs
- 6 Optimisation
- 7 Linear approximations
- 8 Small Variations



L'Hospital's rule

- 1 One of the frequently appearing applications of differentiation is evaluating the so-called **indeterminate type** of limits.

- 2 Recall that

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} x + 1 = 2$$

- 3 This approach is not applicable in all cases. E.g

$$\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$$

- 4 Notice that we can not apply the quotient law of limit as both numerator and denominator have limit 0.
- 5 Limits of the form $\frac{0}{0}$ and $\frac{\infty}{\infty}$ are called **indeterminate type**.
- 6 To evaluate limits of indeterminate type we employ the **L'Hospitals rule**.



L'Hospital's rule: I

Definition

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g = 0$$

or

$$\lim_{x \rightarrow a} f = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g = \pm\infty$$

Then

$$\lim_{x \rightarrow a} \frac{f}{g} = \lim_{x \rightarrow a} \frac{f'}{g'}$$

if the limit on the right side exists (or is $\pm\infty$).

Note: Only apply L'Hospital's rule to indeterminate forms of limits.

Example

Evaluate $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$.

Notice that $\lim_{x \rightarrow 1} \ln x = 0$ and $\lim_{x \rightarrow 1} x - 1 = 0$ hence the given limit is an indeterminate type. Thus

$$\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} = \lim_{x \rightarrow 1} \frac{1/x}{1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1$$

Example

Evaluate $\lim_{x \rightarrow \infty} \frac{e^x}{x}$.

$$\lim_{x \rightarrow \infty} \frac{e^x}{x} = \lim_{x \rightarrow \infty} \frac{e^x}{1} = \lim_{x \rightarrow \infty} e^x = \infty$$



L'Hospital's rule: II

1 Suppose

$$\lim_{x \rightarrow a} f = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g = \pm\infty \quad (1)$$

2 Then $\lim_{x \rightarrow a} f g$ is of indeterminate form

3 if we write

$$\lim_{x \rightarrow a} f g = \lim_{x \rightarrow a} \frac{f}{1/g}$$

4 or

$$\lim_{x \rightarrow a} f g = \lim_{x \rightarrow a} \frac{g}{1/f}$$



Example

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Solution

Notice that this indeterminate type. Thus

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \quad (2)$$

$$= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \quad (3)$$

$$= \lim_{x \rightarrow 0^+} (-x) \quad (4)$$

$$= 0 \quad (5)$$



L'Hospital's rule: III

1 Suppose

$$\lim_{x \rightarrow a} f = \infty \quad \text{and} \quad \lim_{x \rightarrow a} g = \pm\infty \quad (6)$$

2 Then

$$\lim_{x \rightarrow a} (f - g) \quad (7)$$

is indeterminate form of type $\infty - \infty$.

3 We simplify such cases into quotients and apply the L'Hospital's rule.



Example

Evaluate $\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x)$.

Solution

Notice that this indeterminate type. Thus

$$\begin{aligned}
 \lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x) &= \lim_{x \rightarrow (\pi/2)^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) \\
 &= \lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x} \\
 &= \lim_{x \rightarrow (\pi/2)^-} \frac{-\cos x}{-\sin x} \\
 &= 0
 \end{aligned}$$



L'hospital's rule: IV

1 Suppose

$$\lim_{x \rightarrow a} f^g \text{ is} \quad (8)$$

1 0^0 or

2 ∞^0 or

3 1^∞

2 Then it is said to be indeterminate.

3 We solve this by taking the natural logarithm or exponential and solving using the product case.



Example

Evaluate $\lim_{x \rightarrow 0^+} x^x$.

Solution

- 1 Notice that this indeterminate type.
- 2 Let $y = x^x$.
- 3 Then $\ln y = x \ln x$.
- 4 Thus

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = 0 \quad (9)$$

- 5 But $x^x = e^{\ln y}$.
- 6 Hence

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{\ln y} = \lim_{x \rightarrow 0^+} e^{x \ln x} = e^0 = 1$$



Critical Values

- Suppose a function f is continuous on $[a, b]$, then f has some extreme values in $[a, b]$.
- In general, f' may be defined everywhere and possibly zero or may not exist at some points.

Definition (Critical Point)

A **critical Point** of f is a number c such that either

- 1 $f'(c) = 0$ or
- 2 $f'(c)$ does not exist.



Example

Find the critical points of 1). $f(x) = 10x^2 + 4x$ and 2). $f(x) = 6x^5 + 33x^4 - 30x^3 + 100$

We first need the derivative of the function in order to find the critical points

$$f'(x) = 20x + 4 \quad (10)$$

$$20x + 4 = 0 \quad (11)$$

$$x = -\frac{1}{5} \quad (12)$$

$$f'(x) = 30x^4 + 132x^3 - 90x^2 \quad (13)$$

$$= 6x^2(5x^2 + 22x - 15) \quad (14)$$

$$= 6x^2(5x - 3)(x + 5) \quad (15)$$

Equating (15) to zero, then the critical points are

$$x = 0, \quad x = 3/5, \quad x = -5 \quad (16)$$

Minimum and Maximum Values

Our goal is to investigate the maximum and minimum values (**extrema**) of a function. Recall that

Definition

- 1 A function f has an **absolute maximum** (or global maximum) at $\mathbf{c}$ if $f(\mathbf{c}) \geq f(x)$ for all x in D_f . The number $f(\mathbf{c})$ is called the maximum value of on D_f .
- 2 A function f has an **absolute minimum** (or global minimum) at $\mathbf{c}$ if $f(\mathbf{c}) \leq f(x)$ for all x in D_f . The number $f(\mathbf{c})$ is called the minimum value of on D_f .
- 3 A function f has a **local maximum** (or relative maximum) at $\mathbf{c}$ if $f(\mathbf{c}) \geq f(x)$ for all x near $\mathbf{c}$.
- 4 A function f has a **local minimum** (or relative minimum) at $\mathbf{c}$ if $f(\mathbf{c}) \leq f(x)$ for all x near $\mathbf{c}$.



Minimum and Maximum Values

Fermat's theorem

If f has a local maximum(minimum) at c , then c is a critical number of f

the converse of Fermat's Theorem is false in general

To find the extrema of a function f on $[a, b]$

- 1 Find all the critical points of f
- 2 Determine the values of f at the **critical points**, and at a and b .
- 3 Read off the largest and minimum values as absolute maximum and minimum respectively.



Example

Find the absolute maximum(minimum) values of $f(x) = 3x^2 - 12x + 5$ on $[0, 3]$.

The critical points is:

$$f'(x) = 6x - 12 \implies x = 2 \quad (17)$$

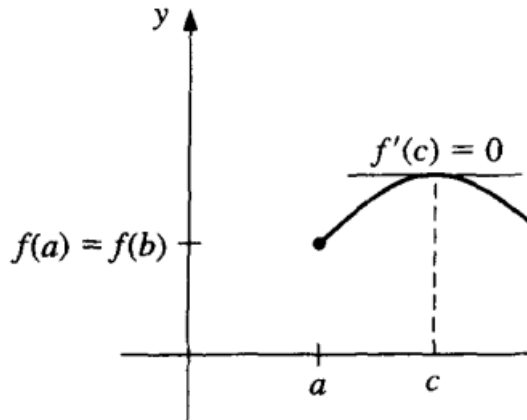
Evaluating

- ① $f(0) = 5$
- ② $f(2) = -7$
- ③ $f(3) = -4$

Therefore, the absolute maximum is 5 and the minimum is -7



Rolle's Theorem



Rolle's theorem

Let f be a function such that

- ① f is continuous on $[a, b]$
- ② f is differentiable on (a, b)
- ③ $f(a) = f(b)$
- ④ Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

This indicates that f has an extrema value at a point of the interval $[a, b]$ if $f(a) = f(b)$.



Mean value theorems

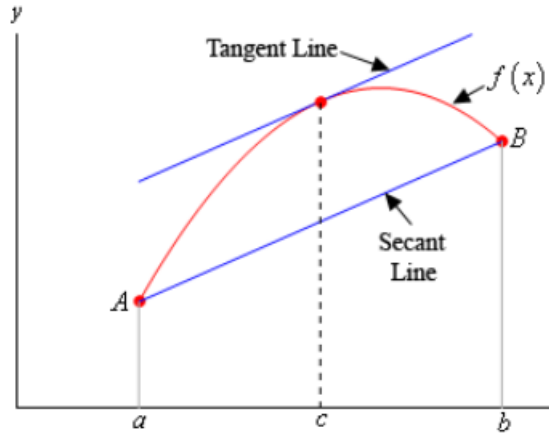
Mean Value Theorem

Let f be a function such that

- ① f is continuous on $[a, b]$
- ② f is differentiable on (a, b)
- ③ Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (18)$$

That is if $f'(c) = m$ then m is the mean value of f on $[a, b]$



Example

Determine all the numbers c which satisfy the conclusions of the Mean Value Theorem for the following function.

$$f(x) = x^3 + 2x^2 - x \quad \text{on} \quad [-1, 2]$$

Since $f(x)$ is a polynomial it is both continuous and differentiable (the derivative exists) on the interval given.

$$f'(x) = 3x^2 + 4x - 1 \quad (19)$$

From the MVT

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)}, \quad 3c^2 + 4c - 1 = \frac{14 - 2}{3}, \quad 3c^2 + 4c - 1 = 4 \quad (20)$$

$$(21)$$

Solving

$$c = 0.7863 \quad \text{and} \quad c = -2.1196 \quad (22)$$

Notice that only one value is in the interval given, hence $c = 0.7863$



Remark

If $f'(x) = 0$ for all x in an interval (a, b) , then $f(x)$ is constant on (a, b) .



Shape of graphs: First Derivative

The derivatives of a function can be used to describe the shape of the graph.

Increasing/Decreasing Test

- 1 If $f'(x) > 0$ on an interval, then f is increasing on that interval.
- 2 If $f'(x) < 0$ on an interval, then f is decreasing on that interval.
- 3 If $f'(x) = 0$ on an interval, then f is constant on that interval.

Example

Determine all intervals where the following function is increasing or decreasing.

$$f(x) = -x^5 + \frac{5}{2}x^4 + \frac{40}{3}x^3 + 5$$



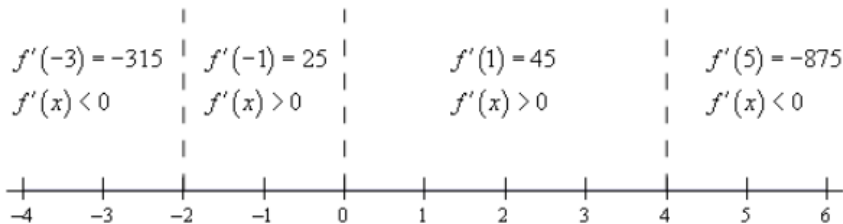
- 1 To determine if the function is increasing or decreasing we will need the derivative.

$$f'(x) = -5x^4 + 10x^3 + 40x^2 \quad (23)$$

$$= -5x^2(x^2 - 2x + 8) \quad (24)$$

$$= -5x^2(x - 4)(x + 2) \quad (25)$$

- 2 We have three critical points: $x = -2$, $x = 0$, $x = 4$



- 3
- 4 So f is increasing on $(-2, 0)$ and $(0, 4)$ and decreasing on $(-\infty, -2)$ and $(4, \infty)$ 

Shape of graphs: First Derivative

These ideas can be summarized up in the following test.

First Derivative Test

Suppose $x = c$ is a critical point of $f(x)$, then

- 1 If $f'(x) > 0$ to the left of $x = c$ and $f'(x) < 0$ to the right of $x = c$ then $x = c$ is a relative maximum
- 2 If $f'(x) < 0$ to the left of $x = c$ and $f'(x) > 0$ to the right of $x = c$ then $x = c$ is a relative minimum
- 3 If $f'(x)$ is the same sign on both sides of $x = c$ then $x = c$ is neither a relative maximum nor a relative minimum.

Note: The first derivative test will only classify critical points as **relative extrema** and not as **absolute extrema**.

Shape of graphs: Second Derivative

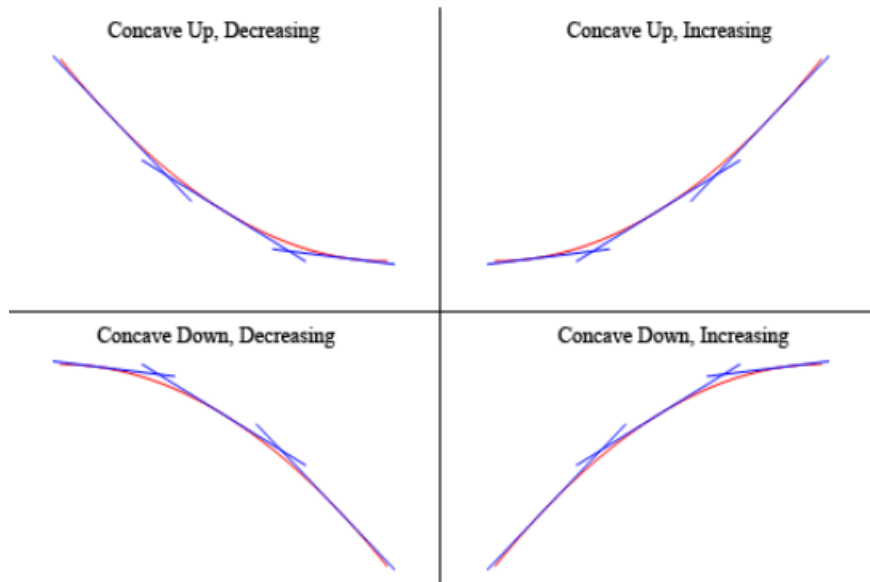
Loosely, a function is **concave up** if it **opens up** and the function is **concave down** if it **opens down**.

Definition

- 1 A function $f(x)$ is **concave up** on an interval I if all of the tangents to the curve on I are below the graph of $f(x)$.
- 2 $f(x)$ is **concave down** on an interval I if all of the tangents to the curve on I are above the graph of $f(x)$.

Note

Concavity has nothing to do with increasing or decreasing. A function can be concave up and either increasing or decreasing. Similarly, a function can be concave down and either increasing or decreasing.



Shape of graphs: Second Derivative

Concavity test

- 1 If $f''(x) > 0$ in $[a, b]$, then the graph of f is concave upward on $[a, b]$.
- 2 If $f''(x) < 0$ in $[a, b]$, then the graph of f is concave downward on $[a, b]$.

Second Derivative Test to classify critical points

Suppose f'' is continuous near a critical point c .

- 1 If $f'(c) = 0$ and $f''(c) > 0$, then f has a relative minimum at c .
- 2 If $f'(c) = 0$ and $f''(c) < 0$, then f has a relative maximum at c .
- 3 If $f'(c) = 0$ and $f''(c) = 0$, then f can have a relative maximum, minimum or neither at c . This is called the **point of inflection**

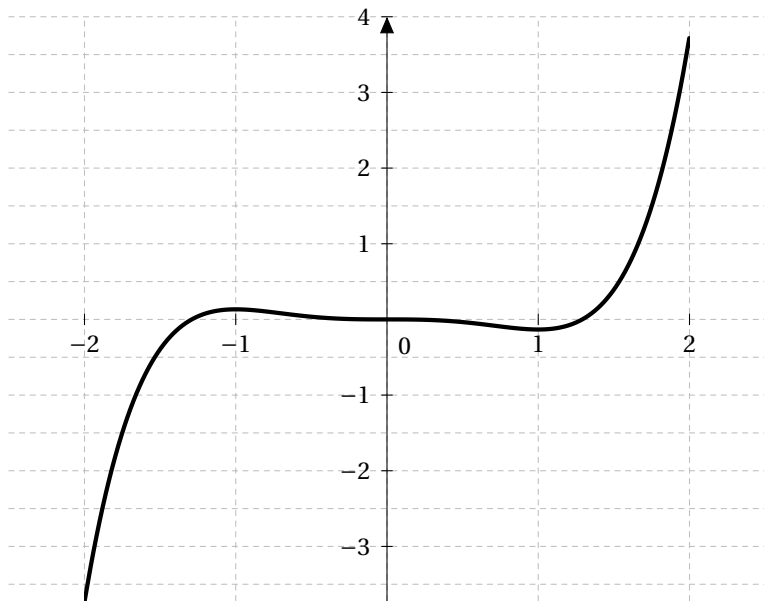


Example

Find the extrema of $f(x) = \frac{1}{5}x^5 - \frac{1}{3}x^3$ on the interval $I = [-2, 2]$

- ① f is continuous and differentiable on $\mathbb{R}$, so on the interval I .
- ② We have $f'(x) = x^4 - x^2 = x^2(x^2 - 1)$ and $f''(x) = 4x^3 - 2x = 2x(2x^2 - 1)$.
- ③ The critical points are: $f'(x) = 0 \implies c = 0$ and $c = \pm 1$.
- ④ Sign of $f''(c)$: $f''(0) = 0$, $f''(-1) = -2 < 0$, $f''(1) = 2 > 0$.
- ⑤ Values of f at critical points and endpoints:
 $f(0) = 0$, $f(\pm 1) = \mp 2/15$, and $f(\pm 2) = \pm 3.733$
- ⑥ Extrema
 - ① f has a global minimum, local maximum, local minimum, and global maximum value at $-2, -1, 1$ and 2 respectively.
 - ② f has an inflection point at $c = 0$.





Optimisation

- 1 In optimization problems we are looking for the largest value or the smallest value that a function can take.
- 2 Here we will be looking for the largest or smallest value of a function subject to some kind of **constraint**.
- 3 The first step in all of these problems should be to very carefully read the problem.
- 4 Once you've done that the next step is to identify the quantity to be optimized and the constraint.
- 5 In identifying the constraint remember that the constraint is the quantity that must be **true** regardless of the solution.



Example

A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

① We seek to maximise the area $A = lb$.

② Subject to the constraint $2b + l = 2400$

Thus $l = 2400 - 2b$ and so

$$A = b(2400 - 2b), \quad 0 \leq b \leq 1200 \quad (26)$$

① Setting $A'(b) = 0$ and solving gives the critical number $b = 600$.

② Finding the absolute maximum is then $A(600) = 720,000$.

③ Hence the dimension is 600×1200 .



Example

We want to construct a box with a square base and we only have $10m^2$ of material to use in construction of the box. Assuming that all the material is used in the construction process determine the maximum volume that the box can have.

- ① We seek to maximise the volume $V = lwh = w^2h$
- ② Subject to the constraint $10 = 2lw + 2wh + 2lh = 2w^2 + 4wh$

Thus $h = \frac{10 - 2w^2}{4w} = \frac{5 - w^2}{2w}$ and so

$$V = w^2 \left(\frac{5 - w^2}{2w} \right) = \frac{1}{2}(5w - w^3) \quad (27)$$

- ① Setting $V'(w) = 0$ and solving gives us the two critical points $w = \pm 1.2910$.
- ② We exclude the negative critical point since we are dealing with length.
- ③ Thus $V(1.2910) = 2.1517m^3$

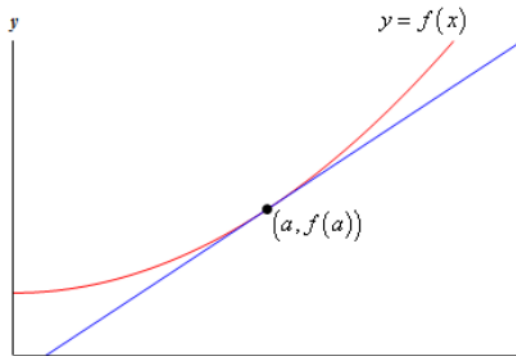


Linear approximations

- 1 Given a function, $f(x)$, we can find its tangent at $x = a$. The equation of the tangent line is given by

$$L(x) = f(a) + f'(a)(x - a) \quad (28)$$

- 2 From this graph we can see that near $x = a$ the tangent line and the function have nearly the same graph.
- 3 On occasion we will use the tangent line, $L(x)$, as an approximation to the function, $f(x)$, near $x = a$.
- 4 In these cases we call the tangent line **the linear approximation** to the function at $x = a$.



Example

Determine the linear approximation for $f(x) = \sqrt[3]{x}$ at $x = 8$. Use the linear approximation to approximate the value of $\sqrt[3]{8.05}$.

$$f'(x) = \frac{1}{3}x^{-2/3}, \quad f(8) = 2, \quad f'(8) = \frac{1}{12} \quad (29)$$

$$L(x) = 2 + \frac{1}{12}(x - 8) = \frac{1}{12}x + \frac{4}{3} \quad (30)$$

$$L(8.05) = 2.00416667 \quad (31)$$

There is no much difference between the exact solution 2.00415802 and its linear approximation 2.00416667.



Small variations

- 1 Consider a function $f(x)$. Changes in x corresponds a change in f .
- 2 Herein, we investigate the rate of change of one quantity in terms of the rate of change of another quantity (which may be more easily measured).
- 3 Typically, we identify the relationship between the quantities and use the chain rule to differentiate both sides with respect to time.



Example

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100\text{cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50cm ?

Let V be the volume of the balloon and let r be its radius. Then

$$\frac{dV}{dt} = 100\text{cm}^3/\text{s}$$

and we seek to find $\frac{dr}{dt}$ when $r = 25\text{cm}$.

Now $V = \frac{4}{3}\pi r^3$. Thus

$$\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Therefore,

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \frac{dV}{dt} = \frac{1}{4\pi(25)^2} 100 = \frac{1}{25\pi}$$



Example

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s , how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

Let x feet be the distance from the bottom of the ladder to the wall and y feet the distance from the top of the ladder to the ground. Then

$$x^2 + y^2 = 100$$

Differentiating gives $\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$. But when $x = 6$, $y = 8$, hence if $\frac{dx}{dt} = 1$, then

$$\frac{dy}{dt} = -\frac{6}{8}(1) = -\frac{3}{4}\text{ ft/s}$$



Exercise

- ① A water tank has the shape of an inverted circular cone with base radius $2m$ and height $4m$. If water is being pumped into the tank at a rate of $2m^3/min$, find the rate at which the water level is rising when the water is $3m$ deep.
- ② A cylindrical can is to be made to hold $1000cm^3$ of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.
- ③ Find the interval within which $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and decreasing.
- ④ Find the critical points, endpoints of D_f , the value of the function at each of these points and identify extreme values. a) $f(x) = x\sqrt{4-x^2}$, b) $f(x) = x^2\sqrt{3-x}$.
- ⑤ The height of a body moving vertically is given by $h(t) = -\frac{1}{2}gt + v_0t + h_0$ where $g > 0$. Find the body's maximum height in terms of h_0, v_0 and g .



Exercise

- 1 Does $f(x) = x^2 - 4x + 1$, $[0, 4]$ satisfy the Rolle's theorem. If yes find c .
- 2 Does $f(x) = \frac{x^2 - x - 6}{x - 1}$, $[-2, 4]$ satisfy the Rolle's theorem. If yes find c .
- 3 Does $f(x) = x^3 - x$, $[0, 2]$ satisfy the Mean Value theorem. If yes find c .
- 4 Use $L(x)$ to linearize $g(x)$ in the following:
 - a) $g(x) = (1 - x)^6$
 - b) $g(x) = \frac{2}{\sqrt{4+3x}}$
 - c) $g(x) = \sqrt[3]{\left(1 - \frac{x}{2+x}\right)^2}$
- 5 The position at time t of a moving particle is given by $s(t)$. Find the time at which its instantaneous velocity $s'(t)$ is equal to its average velocity in the interval T .
 - a) $s(t) = x^2 + 2x - 1$, $T = [0, 1]$. b) $s(t) = t + \frac{1}{t}$, $T = [1/2, 2]$.
 - c) $s(t) = \sqrt{t-1}$, $T = [1, 3]$.



END OF LECTURE
THANK YOU

